Signed-power recovery and inert-prime certificates for ZFC consistency

OpenAI-assisted research for Vladimir Reshetnikov and ProveIt

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Abstract

We test new mechanisms rather than only reoptimize positional weights. The strongest completed polynomial is a sextic in **34 unrestricted integer unknowns**, improving the reviewed 35-variable sextic. Its key step deliberately allows a negative power coordinate, recovers its absolute value by signed Pell orbits, and only then excludes the negative branch by a Mersenne congruence. A cleared transport equation absorbs a quotient without losing integrality. Two positive-domain compositions give degree 30 in 13 unknowns at 94 operations, and degree 114 in 12 unknowns at 105 operations. These are modest numerical improvements, not the large breakthrough sought.

Two other mechanisms are proved at the interface level: inert-prime CRT batching for bounded universal certificates, and a fixed-input boundary absorbed by an existing affine coordinate. Neither has yet been compiled into a smaller complete ordinary polynomial. We give their precise remaining obligations, a failed attempt to merge two Pell roots, and dominated mixed-degree examples so that unsuccessful ideas are not mistaken for records. All consistency claims use explicit inherited arithmetic/compiler results; this is not a new end-to-end proof-assistant certificate.

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1 Objective and what this iteration actually establishes

Let $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{N}_{>0} = \{1, 2, \ldots\}$. Our target is an explicit polynomial P with fixed integer coefficients and a declared domain D such that

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

This is an equivalence, not a proof of either consistency or absence of zeros. The coefficient integers are not chosen depending on the truth of consistency. No unknown is hidden in a coefficient, function call, or register abbreviation. For the three completed constructions, the JSON files specify literal polynomials or arithmetic DAGs, and coefficient bindings identify the full inherited tuple. In contrast, special-function certificates below are explicitly not counted as ordinary polynomials.

There is no single scalar notion of "simplest." We compare the domain, number of unknowns, total degree, live arithmetic-gate count, coefficient size, and proof dependencies. The signed sextic wins one variable, not degree; the positive constructions lower degree at a one-operation cost. No global minimality or first-in-literature claim is made. "New" refers to the mechanism developed in this reviewed iteration, not an exhaustive priority search.

1.1 Immutable comparison scope

The eight incoming archives are pinned by Git blob hash, SHA256, and byte count in `INCOMING_PINS.json`. Arrival was `0c6be9a074d8905b22015e93c5c9d28eda36e95e`; placement was `a2cd79b9120fc75f48dcbef6e38d0f8b18d64d49`. The baseline replay ran all eight original root checkers in fresh copies, without changing the archived sources. Independent live-gate counting covered 44 circuits. The formal-status review was pinned at `9cca523d9f27ad39d52993f4065cbd7e58f8bdd4`. The project lives at

https://github.com/VladimirReshetnikov/ProveIt/tree/main/Computability/DiophantineConsistency.

The input packets are Gap Congruence, Positive Gap, Oriented Guards, Congruence Carriers, Sign and Norm, Sign Recovery, Structural Reductions, and Shared Taps/Carriers. The accompanying original ZIPs and review records make the comparison recoverable without trusting rounded prose metrics.

Domain	Reviewed incoming point	Completed new point	Interpretation
$\mathbb{Z}$ sextic	35 unknowns,20 residuals	34 unknowns,19 residuals	one fewer unknown
$\mathbb{Z}$ quartic	41 unknowns	no improvement	retained incumbent
$\mathbb{Z}$ octic	30 unknowns	30 unknowns	not a record
$\mathbb{N}_{>0}^{13}$	degree 34,93 gates	degree 30,94 gates	degree/cost tradeoff
$\mathbb{N}_{>0}^{12}$	degree 118,104 gates	degree 114,105 gates	degree/cost tradeoff

The incoming nine-positive frontier already includes degree1 494 006/487 and degree1 845 940/444. A late upstream candidate at `bdd86761badd498ca513f1656afba6b4b58ea614` reports the latter degree with 443 operations (450 over naturals), using an even radix and a complete compact54-sharp tuple. It is a reference, not our result; its late replay was not rerun here. Existing ten-integer Sun-based constructions also remain competitive. This iteration does not improve either of those variable-count frontiers, and the new sextic does not invoke Sun's reduction.

1.2 Three distinct levels of assurance

1. Exact finite checks: symbolic identities, degrees, live-gate counts, coefficient hashes, and deliberately corrupted controls.

2. Unbounded mathematical proofs: signed-power recovery, witness reconstruction, CRT soundness and completeness, and conditional boundary transport. These establish more than the finite tests alone can establish.

3. Imported endpoint calibration: the inherited native compiler and Pell/source equivalences connect the arithmetic source to finite ZFC refutations. We identify these dependencies rather than replacing them by an unproved assertion that a universal machine "exists."

None of these statements claims that the new composite endpoint has been accepted by Lean or Rocq. In particular finite tests cannot certify the nonexistence of a counterexample to an unbounded arithmetic lemma.

2 Signed powers and quotient absorption: a 34-integer sextic

The incoming quartic41 system uses both a global square guard $X = 2u_0^2$ and a scale coordinate w_0 satisfying $X = 4qw_0$. A new bootstrap allows us to replace these by the single cubic equation

$$X = 4qu^2.$$

Unlike an ordinary positivity guard, this permits a negative global power at the start. Its sign is recovered only after its absolute value is proved to be a power of two. Simultaneously replace the transport coordinate z by Z and use

$$(K_0q + X)(q - F - \alpha) - qF - Z(q - 1) = 0.$$

Since $q \mid X$ and $\gcd(q, q-1) = 1$, this equation itself forces $q \mid Z$. Thus $z = Z/q$ is recovered without a separate quotient coordinate. The intermediate system has 40 integer variables and 25 residuals.

The new negative-power lemma. All named coefficient ports are the complete source34 tuple, with $q = \beta J + 1$ and $\beta = 2^{1064921} - 1$. The unchanged product guards give $Q = |q| \geq 16$, $Q(Q^2 - 1) < R < Q^4 - 2$, and η, ζ the same nonzero sign. Put $Y = (2v + 1)q^3$, $E = XY$, $a = (X + 1)Y$, $A = a + 2$, $H = 4a+3$, and $k = \eta+\zeta$. The main norm modulo H excludes $u = 0$; hence $|X| \geq 4Q$ and $|E| > 3R+1$.

Let $B = |A|$, $P_0 = |2XY^2 + 1|$, $c_0 = |c|$, and $k_0 = |k|$. The Pell classifications give

$$k_0 = 2\psi_{P_0}(n), \qquad c_0 = \psi_B(p), \qquad p > n \geq 1.$$

The inequality follows from $P_0 > B$ and $c_0 > k_0$. Modulo $|E|$, the first Pell recurrence gives $2n \equiv \epsilon(R+1)$, where $\epsilon = \operatorname{sgn}(k)$ for $X > 0$ and $\epsilon = \operatorname{sgn}(k)(-1)^{n-1}$ for $X < 0$. Either $n \geq (R + 1)/2$, or $n > R$. These bounds justify the unchanged ordinary strong-rank and signed auxiliary step-down lemmas, yielding $p = R$. Consequently the latter branch is impossible and $R = 2n - 1$. This argument precedes all assumptions about the sign of X.

Suppose $X = -x < 0$, and put $Z_0 = |Y|$, $r_0 = (R - 1)/2 \geq 1$. Then $x \geq 64$, $Z_0 \geq 4096$, $B \geq (x - 1)Z_0 - 2$, and $P_0 = 2xZ_0^2 - 1$. Elementary Pell growth gives

$$\frac{c_0}{k_0} \geq \frac{1}{2}\left(\frac{(2B - 1)^2}{2P_0}\right)^{r_0} > \frac{(x - 3)^{r_0}}{2}.$$

The strict comparison follows from the explicit positive difference

$$(2(x - 1)Z_0 - 5)^2 - (x - 3)(4xZ_0^2 - 2) = 4(x + 1)Z_0^2 - 20(x - 1)Z_0 + 25 + 2(x - 3) > 0.$$

Since $c_0/k_0 < Z_0 + 1$, it follows that $Z_0 > 2^R$ and therefore $|H| > x2^R + 1$ and $|H| > x + 2^R$.

For odd R the signed Pell identities modulo H imply, with $\sigma = \operatorname{sgn}(c)$, one of

$$\sigma X \equiv 2^R \pmod{H}, \qquad \sigma X 2^R \equiv -1 \pmod{H}.$$

The strict bounds exclude the second and turn the first into equality. Thus $|X| = 2^R$. But $\operatorname{sgn}(q) = \operatorname{sgn}(X)$, so $q = -2^t$ for some $t \geq 0$. This contradicts $q \equiv 1 \pmod{2^{1064921} - 1}$: the residue of 2^t is a positive power 2^r, $0 \leq r < 1064921$, never -1. Hence $X, q > 0$. The positive branch now gives $\eta, \zeta > 0$ and the inherited positive-power proof recovers $X = 2^R$ and the native compiler endpoint. A literal quartic41 witness is restored by $w_0 = u^2$, $u_0 = 2^{(R-1)/2}$, and $z = Z/q$.

Conversely canonical witnesses have $q = 2^t$ with t odd, R odd, and $R > 3q$. Thus $u = 2^{(R-t-2)/2}$ is integral and $X = 4qu^2$; set $Z = qz$. This proves the same existential equivalence with finite ZFC refutation. It uses the inherited compiler and Pell results, not a new kernel certificate or a proof of consistency.

Low-degree projection. Set $\tau_{\text{old}} = \tau_{\text{tail}} + P - k$ and eliminate

$$P = k + E(c - \eta), \quad F_2 = f^2, \quad g = y^2 - V^2, \quad N = q(L + 1), \quad C_2 = c^2, \quad X = D - ac - \gamma(4a + 3).$$

The first norm becomes $\tau_{\text{tail}}^2 + (2\tau_{\text{tail}} - k)E(c - \eta) - 1$. All 19 remaining residuals are cubic or lower in 34 unrestricted integer unknowns. Their sum of squares has exact degree six: its coefficient at r^2c^4 is four, contributed only by $I - (2r + 1)c^2$. The projections have explicit polynomial inverse reconstructions. This improves the incoming 35-integer sextic by one unknown through a new signed-power bootstrap, not merely through another monic elimination. The straightforward octic companion still has 30 unknowns and is not a new octic variable record.

3 A compact complete specification of the new sextic

This section gives the entire 19-residual equation, not only a recipe for finding it. The four fixed coefficient ports are β, K_0, M_C, M_F, bound as a complete inherited source34 tuple in the JSON artifact and its local upstream provenance. In particular $\beta = 2^{1064921} - 1$; the other ports are supplied by their authenticated coefficient payloads and exact recipes, not arbitrary parameters.

The 34 unrestricted unknowns are

$$J, F, \alpha, f, h, r, v, \tau, \eta, \zeta, y, \gamma, o, j, E, c, a, D, L, V, I, B, \Delta, u, Z,$$
$$s_1, s_2, s_3, t_1, t_2, t_3, w_1, w_2, w_3.$$

Here v is `v_scale`, B is `Baux`, and the three triples are respectively the JSON `FA_pos`, `FC_pos`, and `eta_zeta_pos` coordinates. Define polynomial abbreviations

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha,$$
$$X_* = D - ac - H\gamma, \qquad R_* = [(q - 1)(q - F) + \alpha]L + J(M_C + qM_F),$$
$$\Pi(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + x_3 + 1.$$

The positive-value representation Π is the inherited three-coordinate guard: its range on $\mathbb{Z}^3$ is exactly the positive integers. Its use here requires that precise theorem, not merely that every displayed value

is positive. Let the residuals be

$$
\begin{aligned}
e_1 &= L - q^2 + 1,\\
e_2 &= a - E - (2v+1)q(L+1),\\
e_3 &= E(X_* + 1) - aX_*,\\
e_4 &= c - k(a-E) - \eta,\\
e_5 &= \tau^2 + (2\tau - k)E(c-\eta) - 1,\\
e_6 &= \Delta - a^2 - 4a - 3,\\
e_7 &= D^2 - \Delta c^2 - 1,\\
e_8 &= k - 1 - Eh - R_*,\\
e_9 &= (K_0 q + X_*)C - qF - Z(q-1),\\
e_{10} &= V + c - fo,\\
e_{11} &= V - cj + k - 1 - Eh,\\
e_{12} &= I - (2r+1)c^2,\\
e_{13} &= B - I^2,\\
e_{14} &= B - \Delta f^2 + \Delta,\\
e_{15} &= BV^2 - By^2 + y^2 - 1,\\
e_{16} &= X_* - 4qu^2,\\
e_{17} &= F\alpha - \Pi(s_1, s_2, s_3),\\
e_{18} &= FC - \Pi(t_1, t_2, t_3),\\
e_{19} &= \eta\zeta - \Pi(w_1, w_2, w_3).
\end{aligned}
$$

The requested single equation is

$$\boxed{P_{34} = \sum_{i=1}^{19} e_i^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{34}.}$$

The abbreviations are to be expanded, not existentially quantified. Every residual has degree at most 3, including e_9: X_* has degree2, while C is affine. Hence P_{34} has degree at most6. Its r^2c^4 coefficient is 4, so the degree is exactly 6. Over the integers, $P_{34} = 0$ is equivalent to simultaneous vanishing of all 19 residuals. The prior section's two-way witness construction therefore yields the stated ZFC-provable equivalence, with the inherited endpoint theorem.

The script `check_displayed_system.py` independently compares these compact expressions with all 19 literal JSON rows by symbolic expansion. This prevents an attractive printed abbreviation from silently changing the actual delivered polynomial.

4 Transport absorption meets an adaptive index carrier

The following is a verified incremental composition, not a qualitative breakthrough. It gives thirteen positive variables with degree 30 and 94 arithmetic operations, and twelve positive variables with degree 114 and 105 operations. The respective incoming points are 34/93 and 118/104. The operation increase is explicit.

Use the complete compact source44/source58 coefficient tuple, with $\beta = 2^{590069} - 1$. In the incoming congruence-carrier construction replace only its second carrier by

$$K = 4qY, \quad U = qK, \quad D_* = U(X + ac) + H(U\gamma + KX - r_0), \qquad r_0 = k - R - 1.$$

The first transport carrier, the normalized strong equation, and the auxiliary norm remain unchanged. In particular the thirteen-variable residuals are A_0, A_1, A_a, A_s, with $A_1 = D_*^2 - U^2(1 + \Delta c^2)$, and the equation is $A_0^2 + A_1^2 + A_a^2 + A_s^2 = 0$. Their exact degrees are $15, 13, 13, 14$. The supplied variables and all polynomial registers are listed in the literal JSON graphs; no register introduces an additional witness.

The order of recovery. The unchanged first carrier recovers $q^3 \mid Y$, positive transport, and $C > 0$, before using $q \mid X$. Thus $3q < R < q^4 - 2$ and $\gcd(q, H) = 1$. The new second carrier is positive, and integrality of its square root gives $U \mid H(KX - r_0)$. Since

$$d = \gcd(K, H) \le 3, \qquad \mu = K/d \ge 4q^4/3 > R + 1,$$

we obtain $\mu \mid r_0$. The normalized strong equation and the unchanged signed auxiliary-index argument give $p = R$ without the syntactic shift $X = x + q$ needed by the different ordinary-strong endpoint in the incoming SharedTaps packet. The first Pell congruence then gives $\mu \mid 2n - R - 1$. As $1 \le n < R$, this forces $2n = R + 1$ exactly. Now r_0 is a positive multiple of $4XY^2$, and $U \mid 4XY^2$ because $q^3 \mid Y$. Returning to the full carrier yields $q \mid HX$, hence $q \mid X$. The recovered main-root coordinate $\gamma_{\text{old}} = \gamma + X/q - r_0/U$ is positive because $D > (a + 1)c$ and $c > X$. All native source coordinates are restored.

Completeness and the twelve-variable endpoint. For a canonical source witness take $\gamma = \gamma_{\text{old}} - X/q + r_0/U > 0$; the inherited canonical margin $\gamma_{\text{old}} > X/q$ suffices. Every other witness is unchanged. The incoming divided auxiliary norm still replaces its two coordinates by one and yields the twelve-positive equation $(1 + A_0)(1 + A_1)(1 + N) - 1 = 0$. Its residual degrees are $15, 13, 86$. Because q and U are even, all three residuals are squares modulo four; therefore the unit-product finalizer forces all three residuals to vanish. The exact degree is 114. The corresponding sum-of-squares sibling has unchanged degree 172 and is not presented as a new degree improvement. The ordinary mathematical reduction, with the explicitly inherited compiler and calibration, proves in ZFC that absence of positive zeros is equivalent to Con(ZFC).

A failed additional deletion worth recording. Merging the two outer roots through a rational norm of $\lambda\tau + D$ would remove another witness if it preserved all residue channels. A weight $\lambda = H(q - 1) + 1$ recovers transport but loses $D \equiv X + ac \pmod H$. A weight divisible by H retains that residue but only gives $(q - 1) \mid HB_t$, not $(q - 1) \mid B_t$. These are inequivalent: $q = 32$, $Y = q^3$, and $X = 192$ give $31 \mid H = 4Y(X + 1) + 3$. Imposing $(q - 1) \mid Y$ would repair coprimality but has no justified canonical-completeness proof. Accordingly no eleven-variable endpoint is claimed from this idea. The obstruction is recorded to prevent a superficially attractive but invalid root merger.

The supplied checks verify exact local polynomial identities, degree, liveness, and operation counts. They are not consistency tests and do not replace unbounded proofs or constitute a new proof-assistant certificate.

5 A different certificate mechanism: inert-prime CRT closure

The following is a new compiler-interface theorem within this comparison, not a new measured consistency polynomial. It changes the arithmetic certificate for bounded universal quantification, rather than the source Turing machine or a Jones packing weight. Its complete proof is elementary. The factorial, binomial and variable-exponent graphs have not been expanded and counted in a final ordinary polynomial; that limitation is essential.

Input. Fix $n \ge 1$, natural parameters x, and polynomials $e(i, w; x), f(i, w; x)$ with nonnegative integer coefficients, where $w = (w_1, \ldots, w_m)$ and $m \ge 1$. The target is

$$\forall i < n \; \exists w \in \mathbb{N}^m \quad e(i, w; x) = f(i, w; x).$$

An integer-coefficient polynomial can first be split into positive and negative parts. For a nonempty fixed list, define a binary-tree norm by $\mathcal{N}([z]) = z$ and $\mathcal{N}(L \parallel R) = \mathcal{N}(L)^2 + \mathcal{N}(R)^2$, with both children nonempty.

Explicit certificate. Existentially choose $U, A_1, \ldots, A_m, c \in \mathbb{N}$. Set

$$\begin{aligned} H &= e(n, U, \ldots, U; x) + f(n, U, \ldots, U; x) + n + U + 1, & T &= H!, \\ Q &= (4Tn)^n + 1, & M &= 4TQ + 1, \\ B &= \left((4T)^n n! \tbinom{Q+n}{n}\right) \bmod M, & & \\ z_j &= \tbinom{A_j}{U+1} \quad (1 \le j \le m), & z_{m+1} &= e(c, A; x) - f(c, A; x), \\ z_{m+2} &= 4T(c+1) - 1. & & \end{aligned}$$

Then the target is equivalent to the single condition

$$B \mid \mathcal{N}(z_1, \ldots, z_{m+2}).$$

Here $\binom{A}{k} = 0$ for natural $A < k$. Any future graph replacement must implement that same total convention. These are an explicit special-function certificate, *not yet ordinary polynomial equations.* Their functions can be replaced by supplied Diophantine graphs, but those graph witnesses must be counted.

Product and prime facts. Put $m_i = 4T(i+1) - 1$ for $0 \le i < n$. Then

$$(4T)^n n! \binom{Q+n}{n} = \prod_{j=1}^{n} (4TQ + 4Tj) \equiv \prod_{j=1}^{n} (4Tj - 1) \pmod{M}.$$

The positive product is less than $(4Tn)^n < Q < M$, so $B = \prod_i m_i$. Distinct moduli are coprime: a common divisor divides $j - i$, is coprime to $4T$, and divides $T = H!$ because $j - i \le n \le H$. Each $m_i \equiv 3 \pmod 4$ has a prime divisor $p_i \equiv 3 \pmod 4$. Every prime divisor of every m_i exceeds H, since a prime at most H divides T.

Soundness. Over $\mathbb{F}_p$ for $p \equiv 3 \pmod 4$, the equation $a^2 + b^2 = 0$ forces $a = b = 0$: otherwise it would make -1 a square. Induction through the norm tree makes every leaf zero modulo p_i. The index leaf gives $c \equiv i \pmod{p_i}$. The binomial leaves, multiplied by the invertible $(U+1)!$, give

$$\prod_{r=0}^{U} (A_j - r) \equiv 0 \pmod{p_i}.$$

Primality supplies $w_j \in [0, U]$ congruent to A_j modulo the *same* prime p_i. Hence the local residual at $(i, w; x)$ vanishes modulo p_i. Its absolute value is at most $e(n, U, \ldots, U; x) + f(n, U, \ldots, U; x) < H < p_i$, by monotonicity. It is therefore exactly zero.

Completeness. Choose all finitely many local witnesses, and let U bound them. The CRT supplies $c \equiv i$ and $A_j \equiv w_{ij} \pmod{m_i}$. Each $(U+1)!$ is invertible modulo every m_i, so the binomial leaves vanish modulo m_i; so does the local residual. Thus their norm and the index residual vanish modulo B. This proves both directions without assuming consistency or invoking a standard-model soundness claim. The argument is finitary and formalizable in ZFC.

Why this is different from the incoming constructor. The incoming one-stack source has five local parameters, but its proposed global constructor first introduces 84 nonlinear wires, then applies bounded closure to 89 local witnesses. Apply bounded closure to the original local polynomial instead, and delay degree reduction. Independently, the present theorem replaces separate coordinate bounds by one U, cancels all per-coordinate factorial factors, and combines m range congruences and the evaluation and index congruences into a single norm congruence. At the certificate interface this removes $m-1$ bound coordinates and $m+1$ separate congruence tests. In an implementation using one integer quotient per congruence this also saves $m+1$ quotient slots, but natural-domain graphs may have different bookkeeping. It still needs m binomial graphs. These are not final Diophantine-variable savings; omitted wires can return when the final equation is flattened.

A balanced norm has degree at most $2^{\lceil \log_2(m+2) \rceil}$ in its leaves. Alternatively place the high-degree local residual immediately below the root, so that it is only squared once, and allow more depth for the range leaves. Shared arithmetic DAGs should be preserved throughout: naive substitution into expression trees can overwhelm the savings.

Validation and remaining work. The authored checker verifies 28 exact product/coprimality cases, 2,030 anisotropy pairs, 111 local range/recovery cases and a CRT completeness fixture. Three negative controls demonstrate why ordinary sums of three squares, primes 1 mod 4, and omitted residual bounds are invalid substitutes. These tests supplement rather than prove the theorem. The artifact does not provide a newly kernel-checked closure or a new ordinary-polynomial record. Full graph instantiation, coefficient output, final counts and endpoint calibration remain necessary.

6 A fixed input boundary without a variable-count penalty

This section proves a compiler-interface advance, not a newly instantiated small consistency equation. The unresolved work is an explicit small universal machine, its compiled refuter input, and the new complete coefficient tuple.

The boundary lemma. Suppose the native compiler recovers

$$C = \sum_{i=0}^{N-1} c_i B^i, \qquad 0 \le c_i < B,$$

where B is fixed and increasing i moves leftward from the marked cell. Let $t_0, \dots, t_{m-1}$ be prescribed initializer tiles, encoded with all free dummy digits zero, and let

$$D = \sum_{i=0}^{m-1} \mathrm{enc}(t_i) B^i, \qquad M = B^m.$$

Assume the last digit is nonzero. Then the single linear equation

$$C = D + MV, \qquad V \in \mathbb{Z},$$

is equivalent to prescribing those m digits. Indeed $0 \le D < M$, $C \ge 0$ imply $V \ge 0$, and uniqueness of positional expansion proves both directions. Also $C \ge D \ge B^{m-1}$ forces $N \ge m$. The coefficient M is fixed: no unknown occurs as an exponent.

Affine absorption. The lifted native source has $C = q - F - \alpha$ with $q = \beta J + 1$ and an independent coordinate α. Substitute

$$\boxed{\alpha := q - F - D - MV}$$

everywhere, including in every old positivity guard, and replace the coordinate α by V. There is no new residual. The new solutions are in bijection with the old solutions obeying the boundary constraint, via $V = (C - D)/M$. Affine substitution increases neither variable count nor maximum degree. In particular the existing unrestricted quartic architecture can accept fixed finite inputs without a variable-count or degree penalty. Its coefficients are not the old K215 coefficients: the entire new tuple must be regenerated. The bitlength of M is $1 + \lfloor m \log_2 B \rfloor$ (equal to $1 + m \log_2 B$ when B is a power of two), an explicit cost rather than an uncharged claim of compression.

A constant-size initializer. For an alphabet Γ with blank 0, use horizontal phases L, D, R, vertical reset symbols I, Q, and the tiles

$(L, I, L, 0),\ (R, I, R, 0),$	blank margins,
$(L, I, D, \mathrm{head}(q_0, a)),\ a \in \Gamma,$	first symbol,
$(D, I, D, a),\ a \in \Gamma,$	remaining symbols,
$(D, Q, R, 0),$	marked right delimiter.

For input $a_0 \cdots a_{\ell-1}$, prescribe the delimiter, the data tiles for $a_{\ell-1}, \ldots, a_1$, and the first-symbol tile for a_0, in that right-to-left order. Thus $m = \ell + 1$. An empty input is treated as the single blank symbol. Blank-phase matching and horizontal periodicity force blank margins ending at seams. The ordinary directed-signal transition, finish and reset argument then gives exactly a finite halting computation. The initializer uses $2|\Gamma| + 3$ tiles, independently of input length. Including resets, seams and the impossible forbidden tile, a generous total bound is

$$ns + s(|S_R| + |S_L|) + 6s + 15,$$

for n working states and s symbols. A total five-working-state, five-symbol machine with one extra halt state therefore needs at most 130 tiles. This is a conditional bound, not a pinned 130-tile ZFC instance.

The rank controller survives a frozen prefix. The existing Boolean controller has an odd prime λ, $N = \lambda^n$, $T = N/\lambda$, $C_0 = d\lambda$, $k = \lambda - 1 > 1$, and the orbit identity

$$\{B^{kj} \bmod C_0 T : 0 \le j < T\} = \{1 + C_0 z \bmod C_0 T : 0 \le z < T\}.$$

Retain its coprimality hypotheses and $T > C_0$. Translate its mutable positions to $a + kj$ and the extra position $a + 1$, with fixed $a \ge m$. To represent r, first divide by the unit B^a. Choose $\epsilon \in \{0, 1\}$ so that $rB^{-a} - \epsilon B$ is nonzero modulo λ. Its residue b modulo C_0 satisfies $1 \le b < C_0 < T$ and $\gcd(b, T) = 1$. Write the remaining residue as $b + C_0 u$. Solve

$$bz_0 + \frac{b(b-1)}{2} \equiv u \pmod{T}$$

and select the b orbit coordinates $z_0, \ldots, z_0 + b - 1$, plus the extra position iff $\epsilon = 1$. Their sum is r. They are distinct, and none lies in the frozen prefix. The maximal orbit position is $a + k(T - 1)$, leaving margin $T + k - a$. Arbitrarily large independent padded widths and heights make the required translated positions fit as well. Consequently canonical rank and parity control remains compatible with the fixed input.

The finite replay checks 113,748 positional cases and 108 instances of the translated Boolean controller, as well as 92 finite initialization strips. These checks are not substitutes for the universal argument above. They do not instantiate the missing universal machine, input word, or regenerated arithmetic-controller certificate.

7 Monic projection before positional encoding

A different architecture is to eliminate source registers before compressing source coordinates. The transformed source need not remain quadratic. This yields a rigorously verified positive-domain low-degree tradeoff even before a compact positional code is found.

Let $q = \beta J + 1$. From the frozen 31-coordinate source, eliminate

$$L = q^2 - 1, \quad N = q^3, \quad X = qw, \quad \Delta = a^2 + 4a + 3, \quad C_2 = c^2, \quad F_2 = f^2,$$

$$D = (4a+3)\gamma + ac + X, \quad B_{\mathrm{aux}} = I^2, \quad P = E(c-\eta) + \eta + \zeta, \quad V = jc + hE - \eta - \zeta + 1.$$

The first eight reconstructed coordinates are visibly positive. For the last two use the retained equations $a - E = sN$ and $c = (\eta + \zeta)(a - E) + \eta$, which imply $c > \eta + \zeta$. Each eliminated equation is monic, so substitution and reconstruction give inverse bijections of positive common-zero sets. The exact residual list in `source21.json` consists of 12 polynomials r_j in 21 positive variables, with maximum degree 4 and 100 monomials in total. Thus the explicit equation

$$\sum_{j=1}^{12} r_j^2 = 0$$

is an octic in 21 positive unknowns with the inherited consistency equivalence. The compiler constants are coefficients, not unknowns. For the fixed instance, `COEFFICIENT_BINDING.json` selects the complete inherited old31 tuple with $\beta = 2^{1064921} - 1$, with $K_0 = K$ and the already-shifted M_F. The frozen source hash agrees with the prior package's explicit old31 selection; the separate adaptive-carrier compact44 tuple is not substituted here. This is a mathematical composition with the inherited source theorem, not a new kernel certificate. Deleting only $C_2, F_2, L, X, B_{\mathrm{aux}}$ instead gives 17 cubic residuals in 26 positive unknowns, hence a sextic in 26 positive unknowns. Both are dominated in unknown count at the same degree by existing positive-domain endpoints: a 20-positive octic and a 25-positive sextic. They are source-transformation demonstrations, not new low-degree records. No minimum or improvement over every repository tradeoff is asserted.

For mixed-degree positional compression, let d be the maximum degree, $m = d!$, $W_\alpha = \sum_i \alpha_i w_i$, and $U = \sum_i x_i Z^{w_i}$. Set

$$D_e(Z) = \sum_{j,\,|\alpha|=e} \frac{m}{e!}\Big(\prod_i \alpha_i!\Big) c_{j,\alpha} Z^{q_j - W_\alpha}.$$

Under same-degree isolation against *every* degree-e monomial, the coefficient at q_j in $\sum_e D_e U^e$ equals mr_j by the multinomial theorem. Different degree channels may overlap freely. This removes dummy homogenizing coordinates. However, decoded coordinates bounded by b now require coefficient bounds of order b^d, notb^2. A new code must compensate for that increased radix exponent. The projection alone does not establish a better nine-variable finalizer.

8 What failed, what remains, and the recommended next attack

The signed-power technique is the completed advance: weakening a positivity constraint can lower the number of witnesses if a later arithmetic invariant recovers the sign. Its proof must establish rank bounds *before* assuming that sign. Skipping that ordering would make the argument circular.

The inert-prime certificate changes the bounded-quantifier mechanism itself. The next decisive experiment is not another small exponent search: instantiate its factorial, binomial, power and remainder graphs with shared wires, apply it directly to the small unflattened local refutation relation, and count the entire resulting polynomial. Both soundness and completeness must survive the chosen natural/integer-domain conversions. Until those graphs and all witnesses are explicit, there is no final variable saving to report.

The fixed-input boundary could replace a large specialized transition alphabet by a small universal one. The missing objects are finite but substantial: a pinned machine table with the correct blank-background model, a constructive refuter-to-input compiler, the exact compiled input word, and all regenerated coefficient ports and rank-controller certificates. A theorem of universality by itself is not the required compiler. Input length enters the fixed coefficients and must be charged. No such endpoint is claimed in this packet.

Mixed-degree projection is useful algebra but not a current record: the 26-positive sextic and 21-positive octic are dominated by already archived 25-positive sextic and20-positive octic constructions. Their value here is as checked source alternatives for future positional compression. Higher source degree can worsen the necessary radix exponent enough to erase the benefit of fewer coordinates.

Finally, the attempted 11-variable outer-root merger fails to preserve two independent residue channels simultaneously. The explicit counterexample in `low_degree/FAILED_ROOT_MERGER.md` blocks the proposed cancellation, not every conceivable 11-variable construction. Recording this obstruction is preferable to advertising a root deletion whose converse is unproved.

9 Formalization boundary and reproducibility

The latest reviewed Lean work proves important pieces of signed-index step-down and split-source existence. The signed-index interface admits signed auxiliary congruence coordinates and does not require an odd target index at entry. However our negative-c branch still needs an explicit adapter to that interface. The new polynomial, its literal coefficient binding, native calibration, and the internal ZFC consistency statement have not been newly joined into one checked theorem here. The recent source-polynomial equivalence and counted arithmetic-DAG identity are also different obligations. Repository status and review prose differed on one late build claim, so this report does not silently upgrade it to a verified endpoint. See the pinned formal documents and late snapshot.

The release includes the TeX and PDF, explicit JSON equations and DAGs, authenticated upstream coefficient dependencies, original incoming packets, mathematical proofs, independent audits, and replay scripts. Run `python3 verify.py` from the extracted release. The root checker validates the sealed file manifest and executes the documented component checks in an isolated copy. The recorded validation results distinguish algebraic reconstruction, finite diagnostics, and imported theorems. Run `python3 negative_controls.py` for the packaged integrity controls. No check searches for a ZFC inconsistency or decides consistency.

Sources. All project sources are in Vladimir Reshetnikov's ProveIt repository, https://github.com/VladimirReshetnikov/ProveIt. The archive manifest gives full immutable commits and Git blob IDs; `review/baseline` records comparison evidence, `review/signed34` the signed-power review, `review/positive` the positive/mixed-source review, and `review/interfaces` the two interface audits. The late reference candidate is https://github.com/VladimirReshetnikov/ProveIt/blob/bdd86761badd498ca513f1656afba6b4b58ea614/Computability/DiophantineConsistency/candidates/gap2082-compact54/README.md. These source references support the inherited results and comparison; new mathematical claims are proved in the body and accompanying proof files.